

Chapter 4

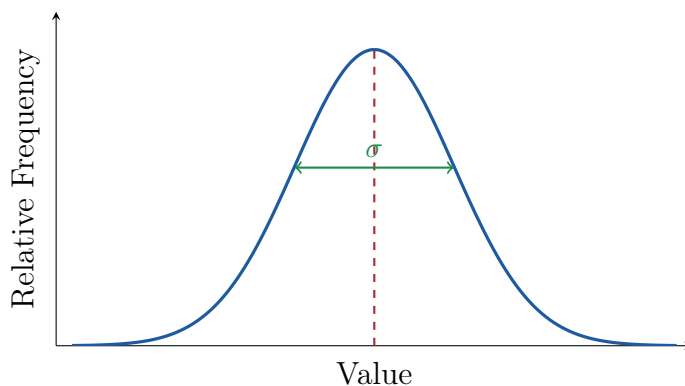
Intermediate Statistics and Inference

The histogram of a large data set often takes a recognizable shape: symmetric, peaked in the center, and tapering toward both ends. This shape, called the normal distribution, is one of the most important structures in statistics. Chapter 3 developed tools for describing data that has already been collected. This chapter extends those tools to the problem of inference: using a sample to draw conclusions about a population the observer cannot fully measure. The z-score places individual values in the context of a distribution. The confidence interval quantifies the uncertainty in an estimate of a population parameter.

4.1 The Normal Distribution

Definition. *Normal distribution.* A **normal distribution** is a continuous, symmetric, bell-shaped probability distribution completely determined by its mean μ and standard deviation σ .

The notation $X \sim N(\mu, \sigma)$ means that the random variable X follows a normal distribution with mean μ and standard deviation σ .



The normal distribution has several properties that follow from its definition.

1. The distribution is symmetric about its mean μ .

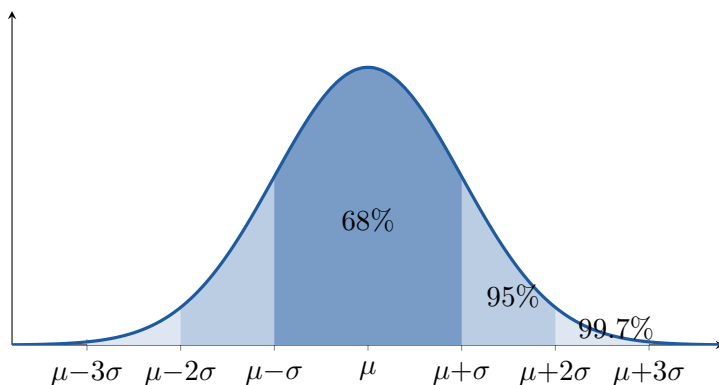
2. The mean, median, and mode are all equal to μ .
3. Approximately 68% of values fall within one standard deviation of the mean.
4. Approximately 95% of values fall within two standard deviations of the mean.
5. Approximately 99.7% of values fall within three standard deviations of the mean.

Properties 3, 4, and 5 together are called the **Empirical Rule**.

Example 4.1.1. Adult heights in a population follow approximately $N(\mu = 67, \sigma = 3)$ inches.

Applying the Empirical Rule:

- About 68% of adults are between 64 and 70 inches tall (within one σ).
- About 95% are between 61 and 73 inches tall (within two σ).
- About 99.7% are between 58 and 76 inches tall (within three σ).



Example 4.1.2. Exam scores for a large class follow $N(\mu = 75, \sigma = 10)$.

- (a) What percentage of students scored between 65 and 85? The interval $[65, 85]$ is $[\mu - \sigma, \mu + \sigma]$. By the Empirical Rule, approximately 68%.
- (b) What percentage scored above 95? The value $95 = \mu + 2\sigma$. The Empirical Rule states 95% of scores fall within two standard deviations, so 5% fall outside. By symmetry, 2.5% fall above 95.
- (c) Between what two scores do the middle 99.7% of students fall? Within three standard deviations of the mean: $[75 - 30, 75 + 30] = [45, 105]$.

Exercises 4.1

4.1.1. The weights of apples in a shipment follow $N(\mu = 180, \sigma = 15)$ grams.

- (a) What percentage of apples weigh between 165 and 195 grams?
- (b) What percentage weigh more than 210 grams?
- (c) Between what two weights do the middle 95% of apples fall?

4.1.2. Commute times for workers in a city follow $N(\mu = 35, \sigma = 8)$ minutes.

- (a) What percentage of commuters travel between 19 and 51 minutes?
- (b) A commuter travels 59 minutes. How many standard deviations above the mean is this commute?
- (c) What percentage of commuters have commutes shorter than 27 minutes?

4.1.3. A normal distribution has mean 50 and standard deviation 6.

- (a) Identify the intervals within one, two, and three standard deviations of the mean.
- (b) What percentage of values fall below 44?
- (c) What percentage of values fall between 38 and 62?

4.2 Z-Scores

Definition. *Z-score.* The **z-score** of a value x from a distribution with mean μ and standard deviation σ is:

$$z = \frac{x - \mu}{\sigma}.$$

The z-score measures how many standard deviations x lies above or below the mean.

A positive z-score indicates a value above the mean. A negative z-score indicates a value below the mean. A z-score of zero indicates a value equal to the mean.

Z-scores allow comparison of values from different distributions. If two observations come from distributions with different means and standard deviations, their raw values cannot be directly compared. Their z-scores can.

Example 4.2.1. Heights follow $N(67, 3)$ inches. What is the z-score for a person who is 73 inches tall?

$$z = \frac{73 - 67}{3} = \frac{6}{3} = 2.0.$$

This person is 2 standard deviations above the mean height. By the Empirical Rule, approximately 97.5% of the population is shorter.

Example 4.2.2. What is the z-score for a height of 61 inches from the same distribution?

$$z = \frac{61 - 67}{3} = \frac{-6}{3} = -2.0.$$

This height is 2 standard deviations below the mean.

Example 4.2.3. (*Recovering the original value*) A distribution has mean $\mu = 50$ and standard deviation $\sigma = 8$. A value has z-score $z = 1.5$. What is the original value?

Solving $z = \frac{x - \mu}{\sigma}$ for x :

$$x = \mu + z \cdot \sigma = 50 + 1.5 \times 8 = 50 + 12 = 62.$$

Example 4.2.4. (*Comparing across distributions*) Student A scores 88 on an exam with mean 80 and standard deviation 10. Student B scores 74 on an exam with mean 65 and standard deviation 6. Which student performed better relative to their class?

$$z_A = \frac{88 - 80}{10} = \frac{8}{10} = 0.80.$$
$$z_B = \frac{74 - 65}{6} = \frac{9}{6} = 1.50.$$

Student B has a higher z-score. Relative to their respective classes, Student B's performance is stronger, despite having a lower raw score.

Exercises 4.2

4.2.1. A distribution has mean $\mu = 100$ and standard deviation $\sigma = 15$. Compute the z-score for each value.

- (a) $x = 115$
- (b) $x = 85$
- (c) $x = 100$
- (d) $x = 130$

4.2.2. A distribution has $\mu = 40$ and $\sigma = 5$. Find the original value x for each z-score.

- (a) $z = 2.0$
- (b) $z = -1.0$
- (c) $z = 0.5$

(d) $z = -3.0$

4.2.3. An athlete runs a mile in 6:40 (400 seconds). The distribution of mile times for the athlete's age group has mean 480 seconds and standard deviation 60 seconds.

- (a) Compute the z-score for this athlete's time.
- (b) Interpret the z-score in one complete sentence.
- (c) Does a negative z-score indicate a good or poor performance for a running time? Explain.

4.2.4. Two students take standardized tests in different subjects. Student A scores 520 on a test with mean 500 and standard deviation 100. Student B scores 28 on a test with mean 21 and standard deviation 5.

- (a) Compute the z-score for each student.
- (b) Which student performed better relative to their group? Justify.

4.3 Sampling and Estimation

Chapter 2 introduced the distinction between a population and a sample. The goal of statistical inference is to use measurements from a sample to estimate unknown properties of the population.

Definition. *Population parameter.* A **population parameter** is a numerical property of the entire population, such as the population mean μ or the population standard deviation σ . Parameters are usually unknown.

Definition. *Sample statistic.* A **sample statistic** is a numerical property computed from the sample, such as the sample mean $\bar{x}$ or the sample standard deviation s . Statistics are calculated from observed data and used to estimate parameters.

The sample mean $\bar{x}$ is the primary estimate of the population mean μ . Different samples from the same population produce different values of $\bar{x}$. This variability is called **sampling variability**. It does not indicate an error in the process. It is an inherent property of sampling.

Example 4.3.1. A manufacturer wants to estimate the mean battery life of a new phone model. Testing every phone in production is impractical. Instead, a sample of 100 phones is tested. The sample mean battery life is $\bar{x} = 18.4$ hours.

The population mean μ is unknown. The sample mean $\bar{x} = 18.4$ hours is the best available estimate of μ . A different sample of 100 phones would likely produce a slightly different value of $\bar{x}$.

Exercises 4.3

4.3.1. For each situation, identify the population parameter being estimated and the sample statistic used to estimate it.

- (a) A polling organization surveys 1,200 voters and finds that 58% support a candidate.
- (b) A biologist measures the lengths of 40 randomly caught fish and finds a mean length of 31.2 cm.
- (c) A food scientist tests 50 cereal boxes and finds a mean weight of 368 grams.

4.3.2. Explain in two complete sentences why taking a larger sample generally produces a more reliable estimate of the population mean, even though the sample mean will still differ from the population mean.

4.3.3. A school district wants to estimate the mean number of books read per student per year. Describe how you would select a simple random sample of 100 students from a school of 800 students.

4.4 Confidence Intervals

A sample mean $\bar{x}$ is a point estimate of the population mean μ . A **confidence interval** extends this to an interval estimate: a range of plausible values for μ .

Definition. *Confidence interval.* A **confidence interval** for a population mean is an interval of the form:

$$\bar{x} - E \text{ to } \bar{x} + E,$$

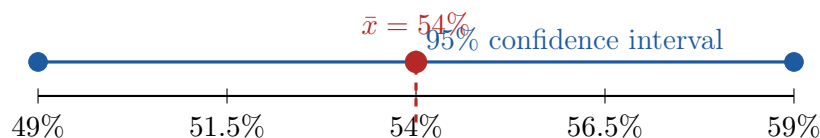
where E is the **margin of error**. A 95% confidence interval means: if the same sampling process were repeated many times, approximately 95% of the resulting intervals would contain the true population mean μ .

The margin of error E depends on the sample size and the variability in the data. Larger samples produce smaller margins of error. More variable data produces larger margins of error.

Example 4.4.1. A survey of 400 registered voters finds that 54% plan to vote yes on a ballot measure. The margin of error is $\pm 5\%$.

The 95% confidence interval for the true proportion supporting the measure is:

$$(54 - 5)\% \text{ to } (54 + 5)\% = 49\% \text{ to } 59\%.$$



The confidence interval includes values below 50%, so the data does not rule out the possibility that fewer than half of all voters support the measure.

Example 4.4.2. A researcher samples 100 students and finds their mean daily screen time is $\bar{x} = 4.2$ hours with a margin of error of ± 0.3 hours.

The 95% confidence interval is:

$$4.2 - 0.3 \text{ to } 4.2 + 0.3 = 3.9 \text{ to } 4.5 \text{ hours.}$$

A policy maker who wants to know whether students average more than 4 hours of screen time per day can use this interval. Since the entire interval lies above 4 hours, the data supports the conclusion that the population mean exceeds 4 hours.

Example 4.4.3. (*Effect of sample size*) Two surveys estimate the same population proportion. Survey A samples 100 people and reports a margin of error of $\pm 10\%$. Survey B samples 1,000 people and reports a margin of error of $\pm 3\%$.

Both surveys find a sample proportion of 62%.

	Survey A	Survey B
Sample size	100	1,000
Sample proportion	62%	62%
Margin of error	$\pm 10\%$	$\pm 3\%$
Confidence interval	52% to 72%	59% to 65%

Survey B provides a much narrower interval and a more precise estimate of the population proportion.

Exercises 4.4

4.4.1. Construct the 95% confidence interval for each situation.

- (a) $\bar{x} = 82$, margin of error = 4
- (b) $\bar{x} = 3.7$ hours, margin of error = 0.5 hours
- (c) $\bar{x} = 47\%$, margin of error = 6%

4.4.2. A poll finds 38% support for a proposal with a margin of error of $\pm 4\%$.

- (a) State the confidence interval.
- (b) A politician claims more than 40% of the population supports the proposal. Is this consistent with the confidence interval? Explain.

4.4.3. A nutrition label states that a serving of cereal contains a mean of 230 calories. A consumer group tests 64 boxes and finds $\bar{x} = 243$ calories with a margin of error of ± 8 calories.

- (a) State the confidence interval.
- (b) Does the interval support or contradict the label's claim of 230 calories? Explain in two sentences.

4.4.4. A researcher wants to reduce their margin of error by half. If the original sample size was 200, approximately what sample size is needed? (Note: the margin of error is approximately proportional to $\frac{1}{\sqrt{n}}$, where n is the sample size.)

4.5 The Observer and the Estimate

Statistical inference is fundamentally an act of observation under constraint. The population is the complete object of study. The sample is what an observer can access. Every estimate produced from a sample is shaped by two things: the question the observer is asking, and the resolution at which the observation is made.

A z-score asks: how unusual is this particular value relative to the distribution? A confidence interval asks: what range of population values is consistent with the observed sample? Both questions are asked of the same data. They produce different answers because they are different questions.

Tool	Question	Output
Z-score	How far is this value from the mean?	A number in units of σ
Empirical Rule	What fraction falls within $k\sigma$?	A percentage
Confidence interval	Where is the population mean likely?	An interval

Choosing the wrong tool for the question produces a technically correct calculation that does not answer the question being asked. Precision in the question determines the value of the answer.

Example 4.5.1. A health researcher is studying resting heart rates. The sample mean is $\bar{x} = 72$ beats per minute with a margin of error of ± 3 bpm. A particular subject has a resting heart rate of 91 bpm. The population distribution has $\sigma = 10$ bpm.

The researcher has two questions.

Question 1. Is this subject's heart rate unusual?

$$z = \frac{91 - 72}{10} = 1.9.$$

The subject is 1.9 standard deviations above the mean. By the Empirical Rule, fewer than 3% of the population has a heart rate this high or higher.

Question 2. What is the plausible range for the true population mean heart rate?

$$72 - 3 \text{ to } 72 + 3 = 69 \text{ to } 75 \text{ bpm.}$$

The z-score answers a question about an individual. The confidence interval answers a question about the population. Neither answer substitutes for the other.

Exercises 4.5

4.5.1. A data set has mean $\mu = 200$ and standard deviation $\sigma = 25$. An observation of $\bar{x} = 212$ is recorded from a sample of 50. The margin of error is ± 7 .

- (a) Compute the z-score for a single value of $x = 245$.
- (b) Construct the confidence interval for the population mean.
- (c) Explain in two sentences why the z-score and the confidence interval answer different questions.

4.5.2. A distribution has mean $\mu = 60$ and standard deviation $\sigma = 12$.

- (a) A value of 78 is observed. Is this within two standard deviations of the mean?
- (b) A sample of 80 observations produces $\bar{x} = 63$ with margin of error ± 3 . Does the confidence interval contain $\mu = 60$?

Chapter 4 Review

5. The lifetimes of a brand of light bulb follow $N(\mu = 1,200, \sigma = 100)$ hours.

- (a) What percentage of bulbs last between 1,000 and 1,400 hours?
- (b) What percentage last more than 1,400 hours?
- (c) A bulb lasts 940 hours. Compute its z-score and interpret it in one sentence.
- (d) Between what two lifetimes do the middle 68% of bulbs fall?

5. A medical study measures blood pressure for a sample of 150 adults and finds $\bar{x} = 122$ mmHg with a margin of error of ± 5 mmHg.
 - (a) Construct the 95% confidence interval.
 - (b) A medical guideline states that a healthy population mean is below 120 mmHg. Is this consistent with the confidence interval?
 - (c) If the sample size were increased to 600, would the margin of error increase or decrease? Explain.
5. Two job applicants take aptitude tests from different testing companies. Applicant A scores 78 on a test with mean 70 and standard deviation 8. Applicant B scores 65 on a test with mean 55 and standard deviation 6.
 - (a) Compute the z-score for each applicant.
 - (b) Which applicant performed better relative to their respective test population?
 - (c) If the employer only sees the raw scores, what conclusion might they incorrectly draw?
5. A distribution follows $N(\mu = 500, \sigma = 80)$.
 - (a) A value has z-score $z = -0.75$. What is the original value?
 - (b) What percentage of values fall below $\mu - 2\sigma$?
 - (c) What is the z-score for the value $\mu + 3\sigma$?
5. A poll of 900 adults finds that 43% approve of a new city policy, with a margin of error of $\pm 3\%$.
 - (a) State the confidence interval.
 - (b) A local official claims a majority of residents approve. Is this consistent with the interval?
 - (c) Another poll of 100 adults finds the same sample proportion of 43% but reports a margin of error of $\pm 10\%$. Which poll provides a more precise estimate? Explain.
5. Explain in three complete sentences the difference between a population parameter and a sample statistic. Use a specific example to illustrate.
5. A researcher measures the z-score of a single subject and finds $z = 2.5$. The population has mean $\mu = 80$ and standard deviation $\sigma = 6$.
 - (a) What is the subject's original value?
 - (b) Using the Empirical Rule, approximately what percentage of the population has a higher value than this subject?